Deepak Sathyan

Publication List

Publications

Author order is alphabetical by last name, as per HEP convention.

6. New Constraints on Neutrino-Dark Matter Interactions: A Comprehensive Analysis — July 2025
P.S.B. Dev, D. Kim, **D. Sathyan**, K. Sinha, Y. Zhang
arxiv.org/abs/2507.01000 (JHEP 10 (2026) 207)

5. New laboratory constraints on neutrinophilic mediators — July 2024
P.S.B. Dev, D. Kim, **D. Sathyan**, K. Sinha, Y. Zhang
arxiv.org/abs/2407.12738 (Phys.Lett.B 868 (2025) 139765)

4. 'Unification' of BSM searches and SM measurements: the case of lepton+MET and m_W — Apr 2024
K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, **D. Sathyan**
arxiv.org/abs/2404.17574 (JHEP 02 (2025) 139)

3. A new purpose for the W-boson mass measurement: Searching for New Physics in lepton+MET — Oct 2023
K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, **D. Sathyan**
arxiv.org/abs/2310.13687 (Phys.Lett.B 855 (2024) 138774)

2. Energy-peak based method to measure top quark mass via B-hadron decay lengths — Dec 2022
K. Agashe, S. Airen, R. Franceschini, J. Incandela, D. Kim, **D. Sathyan**
arxiv.org/abs/2212.03929 (JHEP 06 (2023) 021)

1. LHC Signals for KK Graviton from an Extended Warped Extra Dimension — Aug 2020
K. Agashe, M. Ekhterachian, D. Kim, **D. Sathyan**
arxiv.org/abs/2008.06480 (JHEP 11 (2020) 109)

Preprints

3. Intensity-Frontier Signals of Warped Extra Dimensions — Aug 2026
D. Kim, **D. Sathyan**, A. Verma
arxiv.org/abs/2608.20482

2. Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering — May 2026
B. Dutta, D. Goswami, J. Kumar, M. Rai, **D. Sathyan**
arxiv.org/abs/2605.08010

1. A Baryon and Lepton Number Violation Model Testable at the LHC — Aug 2025
A. Bhoonah, F. Burk, D. Liu, T. Ou, **D. Sathyan**
arxiv.org/abs/2508.21064

Articles in Preparation

1. **LHC as a Beam Dump: Probing Light Dark Photons**
   B. Dutta, A. Karthikeyan, D. Kim, H. Kim, T. Kim, **D. Sathyan**
   (In preparation)

2. **New Probes of Dark Sectors at a Muon Collider**
   B. Dutta, A. Karthikeyan, D. Kim, K. Kelly, **D. Sathyan**
   (In preparation)

3. **Exploring open systems approach to neutrino oscillations**
   A. Chandra Shekar, K. Kelly, **D. Sathyan**, L. Strigari, T. Zhou
   (In preparation)

4. **LHC Signals for KK Gravitons Producing 8-Particle Final States**
   K. Agashe, D. Kim, P. Maksimovic, S. Mondal, M. Osherson, K. Panchal, D. Plotnikov, **D. Sathyan**
   (In preparation)

Snowmass2021 Contributions

5. **TF07 Snowmass Report: Theory of Collider Phenomena** — Oct 2022
   F. Maltoni et al.
   arxiv.org/abs/2210.02591

4. **Report of the Topical Group on Top quark physics and heavy flavor production for Snowmass 2021** — Sept 2022
   K. Agashe et al.
   arxiv.org/abs/2209.11267

3. **Report of the Topical Group on Physics Beyond the Standard Model at Energy Frontier for Snowmass 2021** — Sept 2022
   T. Bose et al.
   arxiv.org/abs/2209.13128

2. **Snowmass2021 - White Paper, Implications of Energy Peak for Collider Phenomenology: Top Quark Mass Determination and Beyond** — Apr 2022
   K. Agashe, S. Airen, R. Franceschini, D. Kim, **D. Sathyan**
   arxiv.org/abs/2204.02928

1. **Snowmass2021 White Paper: Collider Physics Opportunities of Extended Warped Extra-Dimensional Models** — Mar 2022
   K. Agashe, J.H. Collins, P. Du, M. Ekhterachian, S. Hong, D. Kim, R.K. Mishra, **D. Sathyan**
   arxiv.org/abs/2203.13305

Muon g-2 Articles

3. **The straw tracking detector for the Fermilab Muon g-2 Experiment** — Nov 2021
   B.T. King et al.
   arxiv.org/abs/2111.02076 (JINST 17 (2022) P02035)

2. **Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab** Apr 2021
   Muon g-2 Collaboration
   arxiv.org/abs/2104.03240 (Phys.Rev.Accel.Beams 24 (2021) 044002)

1. **Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm** Apr 2021
   Muon g-2 Collaboration
   arxiv.org/abs/2104.03281 (Phys.Rev.Lett. 126 (2021) 141801)

Ph.D. Thesis

1. **Unifying Searches for New Physics with Precision Measurements of the W Boson Mass** 2024
   D. Sathyan
   drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9 (Ph.D. Thesis)